COER UNIVERSITY

ASSIGNMENT-1

Course: MCA | Subject:MachineLearning (MCA201T)

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Q1. Define Machine Learning and explain how it differs from conventional rule-based programming. Discuss the role of data, algorithms, and models in a machine learning system.

Definition

Machine Learning (ML) is a branch of Artificial Intelligence in which computer systems learn patterns from data and improve their performance on a task with experience, without being explicitly programmed with fixed rules for every situation. Arthur Samuel (1959) described it as the field of study that gives computers the ability to learn without being explicitly programmed. A more formal definition by Tom Mitchell (1997) states: a computer program is said to learn from experience E with respect to some task T and performance measure P, if its performance at T, as measured by P, improves with experience E.

Example: In spam filtering, T = classifying emails as spam or not spam, E = a large set of emails labelled by users, and P = the percentage of emails classified correctly.

Difference from Conventional Rule-Based Programming

Basis	***Conventional (Rule-Based) Programming***	***Machine Learning***
Input to the system	Data + hand-written rules (program).	Data + expected answers (labels), for supervised learning.
Output	Answers produced by applying the rules.	Rules / a model learned from the data, which is then used to produce answers.
Who defines the logic	The programmer.	The algorithm discovers the logic from data.
Handling complexity	Difficult when rules are too many or unknown (e.g., face recognition).	Well suited to complex problems where rules cannot be written easily.
Adaptability	Must be manually rewritten when conditions change.	Can be retrained on new data to adapt.
Nature of results	Deterministic and exact.	Probabilistic; results have some error.
Dependence on data	Low.	Very high; quality of data decides quality of the model.
Example	Payroll calculation, tax computation.	Spam detection, image recognition, recommendations.

Role of Data, Algorithms and Models

- ***Data:*** Data is the experience from which the system learns. It consists of examples described by features (inputs) and, in supervised learning, labels (targets). The quantity, quality, diversity and representativeness of data decide how well the system can learn. Data is normally divided into training, validation and test sets.
- ***Algorithm:*** An algorithm is the learning procedure (such as linear regression, decision tree, k-means, neural network trained with gradient descent) that examines the data and adjusts internal parameters to minimise a loss function. It defines ***how*** learning takes place.

• ***Model:*** A model is the output of the learning process: the mathematical function with learned parameters (e.g., weights of a network or split rules of a tree) that maps new inputs to predictions. Once trained, the model is used for inference on unseen data.

In short: ***Data + Algorithm -> (training) -> Model; New data + Model -> Predictions.*** Data provides the knowledge, the algorithm is the method of learning, and the model is the learned knowledge that is finally deployed.

Q2. Explain supervised, unsupervised, and reinforcement learning. Compare their learning objectives, type of feedback, common algorithms, and suitable application scenarios.

1. Supervised Learning

The model learns from a ***labelled dataset*** in which every input has a known correct output. The goal is to learn a mapping from inputs to outputs so that the model can predict the output for new inputs. It is divided into ***classification*** (discrete output, e.g., spam / not spam) and ***regression*** (continuous output, e.g., house price).

2. Unsupervised Learning

The model is given ***unlabelled data*** and must discover hidden structure, patterns or groupings on its own. Typical tasks are clustering (grouping similar items), dimensionality reduction, association rule mining and anomaly detection.

3. Reinforcement Learning (RL)

An ***agent*** interacts with an ***environment***, takes actions, and receives ***rewards or penalties***. The agent learns a ***policy*** that maximises the cumulative reward over time, through trial and error. Learning involves the balance between exploration (trying new actions) and exploitation (using known good actions).

Comparison

Basis	***Supervised***	***Unsupervised***	***Reinforcement***
Learning objective	Learn a mapping from inputs to known outputs; predict labels or values.	Discover structure, groups or patterns in data.	Learn a policy that maximises long-term cumulative reward.
Type of feedback	Direct feedback: correct label for every example.	No feedback / no labels.	Delayed, evaluative feedback in the form of reward or penalty.

Basis	*Supervised*	*Unsupervised*	*Reinforcement*
Data	Labelled data.	Unlabelled data.	Experience generated by interaction with an environment.
Common algorithms	Linear and logistic regression, decision trees, random forest, SVM, k-NN, Naive Bayes, neural networks.	k-means, hierarchical clustering, DBSCAN, PCA, Apriori, autoencoders.	Q-learning, SARSA, Deep Q-Network (DQN), policy gradient, actor-critic.
Application scenarios	Spam detection, disease diagnosis, credit scoring, price prediction.	Customer segmentation, market basket analysis, anomaly detection, topic discovery.	Game playing (AlphaGo), robotics, self-driving, dynamic pricing, resource scheduling.
Evaluation	Accuracy, precision, recall, RMSE, etc.	Silhouette score, inertia, domain judgement.	Cumulative reward, convergence of policy.

Q3. Discuss the major real-world applications of Machine Learning in healthcare, finance, education, cybersecurity, recommendation systems, and transportation. Explain one application in detail.

Domain	*Major Applications*
Healthcare	Disease detection from medical images (e.g., diabetic retinopathy, tumours in X-ray/MRI), early prediction of diseases, patient risk and readmission prediction, drug discovery, personalised treatment, and analysis of electronic health records.
Finance	Credit card fraud detection, credit scoring and loan approval, algorithmic trading, risk management, customer churn prediction, insurance claim analysis, and chatbots for customer service.
Education	Predicting student performance and dropout, personalised learning paths, intelligent tutoring systems, automated grading, plagiarism detection, and adaptive quizzes.
Cybersecurity	Intrusion and malware detection, phishing and spam email filtering, anomaly detection in network traffic, user behaviour analytics, and automated threat response.
Recommendation systems	Product suggestions on e-commerce sites, movie and video recommendations (Netflix, YouTube), music playlists (Spotify), news feed ranking, and targeted advertisements using collaborative and content-based filtering.
Transportation	Self-driving vehicles, traffic flow prediction and signal optimisation, route planning and ETA estimation (map apps), ride demand and dynamic pricing, predictive maintenance of vehicles, and logistics optimisation.

Detailed Explanation: Credit Card Fraud Detection (Finance)

Problem: Banks must identify fraudulent card transactions in real time (within milliseconds) among millions of legitimate ones, without blocking genuine customers.

1. ***Data:*** Historical transactions with features such as amount, time, merchant category, location, device, frequency of recent transactions, distance from the previous transaction, and the customer's usual spending pattern. Each past transaction is labelled fraud or genuine (from customer complaints and investigations).

2. ***Learning type:*** Supervised binary classification (fraud = 1, genuine = 0). Unsupervised anomaly detection (e.g., isolation forest, autoencoders) is used in addition to catch new fraud patterns for which no labels exist.

3. ***Challenge - class imbalance:*** Fraud is typically far below 1% of transactions, so accuracy is misleading. Techniques such as SMOTE / oversampling, under sampling, class weights and cost-sensitive learning are used.

4. ***Models:*** Logistic regression (baseline), random forest, gradient boosting (XGBoost / LightGBM), and neural networks.

5. ***Evaluation:*** Precision, recall, F1-score, and area under the precision-recall curve, because missing a fraud (false negative) and blocking a genuine customer (false positive) both have costs.

6. ***Deployment:*** The model scores each transaction in real time. High-risk transactions are declined or sent for OTP / manual verification. The model is monitored and retrained regularly, since fraudsters constantly change their methods (concept drift).

7. ***Benefit:*** Reduced financial loss, faster detection, fewer false alarms and increased customer trust.

Q4. Explain the major stages involved in designing a Machine Learning system, starting from problem formulation and data collection to model evaluation and deployment. Why is problem definition important?

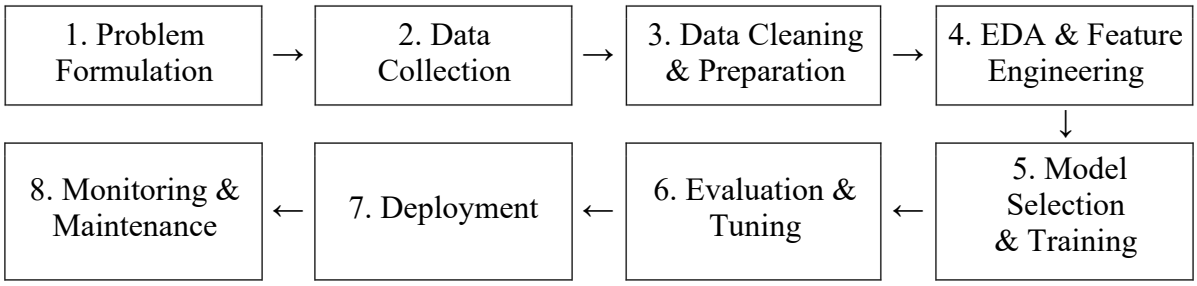


Figure: Stages of a Machine Learning system (iterative process; results of later stages often send the team back to earlier ones)

1. ***Problem formulation:*** Understand the business or research goal, decide what is to be predicted, whether ML is really needed, the type of task (classification, regression, clustering, etc.), the success metric, constraints (time, cost, latency, privacy) and available resources.

2. ***Data collection:*** Identify and gather relevant data from databases, logs, sensors, surveys, APIs or public datasets. The data must be sufficient, relevant, representative and legally usable.

3. ***Data cleaning and preparation:*** Handle missing values, duplicates, outliers and inconsistent formats; encode categorical variables; normalise or scale features; and split the data into training, validation and test sets.

4. ***Exploratory data analysis (EDA) and feature engineering:*** Visualise and summarise the data to understand distributions and relationships; create, transform and select the most useful features; reduce dimensions if needed.

5. ***Model selection and training:*** Choose suitable algorithms (starting with a simple baseline), train them on the training data, and tune hyperparameters using the validation set or cross-validation.

6. ***Model evaluation:*** Test the final model on unseen test data using appropriate metrics (accuracy, precision, recall, F1, ROC-AUC for classification; MAE, RMSE, R^2 for regression). Check for overfitting, bias and fairness, and compare against the baseline and business requirements.

7. ***Deployment:*** Integrate the model into a real application (web service / API, mobile app, batch pipeline or edge device) with proper scalability, security and versioning.

8. ***Monitoring and maintenance:*** Track performance, data drift and concept drift in production; collect feedback; retrain and update the model when performance falls.

Why Problem Definition is Important

- It gives a ***clear direction***: the wrong problem can lead to an excellent model that solves nothing useful.
- It decides ***what data*** is needed, what the target variable is, and which algorithms and metrics are suitable.
- It links technical work with ***business value*** and sets measurable success criteria, so that results can be judged objectively.
- It identifies ***constraints and risks*** (ethical, legal, cost, time) early, and prevents waste of time and money on rework.
- It helps decide whether a simple rule-based or statistical solution is enough and ML is not required.

Q5. Trace the historical development of Machine Learning from early computational learning ideas to modern deep learning. Mention significant milestones and explain how improvements in data and computing influenced the field.

Period	***Milestones***
1940s-1950s (Foundations)	1943: McCulloch and Pitts proposed a mathematical model of the artificial neuron. 1950: Alan Turing published "Computing Machinery and Intelligence" and proposed the Turing Test. 1952: Arthur Samuel wrote a self-improving checkers program; he coined the term "machine learning" in 1959. 1956: The Dartmouth Conference established AI as a research field. 1957: Frank Rosenblatt invented the Perceptron.

Period	*Milestones*
1960s-1970s (Early setbacks)	1967: The nearest neighbour algorithm was proposed. 1969: Minsky and Papert showed limitations of single-layer perceptrons (they cannot solve XOR), which reduced funding and led to the first "AI winter". Research shifted towards symbolic AI and expert systems.
1980s (Revival)	Decision-tree algorithms (ID3 by Quinlan, CART) became popular. 1986: Rumelhart, Hinton and Williams popularised the backpropagation algorithm for training multi-layer neural networks. 1989: Yann LeCun applied convolutional networks to handwritten digit recognition; Watkins introduced Q-learning.
1990s-2000s (Statistical ML)	Focus moved to statistical, data-driven methods. 1995: Support Vector Machines (Cortes and Vapnik). 1997: LSTM networks (Hochreiter and Schmidhuber); IBM Deep Blue defeated Kasparov. 2001: Random Forests (Breiman). Boosting methods such as AdaBoost, and ensemble learning, became widely used. 2006: Hinton and colleagues showed effective training of deep belief networks, re-introducing "deep learning".
2009-2015 (Deep learning breakthrough)	2009: ImageNet dataset was released. 2012: AlexNet, trained on GPUs, won the ImageNet challenge by a large margin. 2014: Generative Adversarial Networks (Goodfellow et al.). 2015: ResNet enabled very deep networks. 2016: DeepMind AlphaGo defeated the world Go champion Lee Sedol using deep reinforcement learning.
2017-present (Transformers and generative AI)	2017: The Transformer architecture ("Attention Is All You Need"). 2018: BERT and GPT models. 2020 onwards: Large language models (GPT-3 and successors), diffusion models for image generation, and foundation models; wide use of chat assistants, code generation and multimodal systems.

Influence of Data and Computing

- ***Big data:*** The growth of the internet, social media, smartphones, sensors and digital records created huge labelled and unlabelled datasets (such as ImageNet). Deep models need large data to generalise well, so data availability became a main driver of progress.
- ***Computing power:*** Faster CPUs, and above all GPUs and later TPUs, made it possible to train large neural networks in days instead of years. Cloud computing gave affordable access to this hardware.
- ***Algorithmic advances:*** Better activation functions (ReLU), optimisers (Adam), regularisation (dropout, batch normalisation) and architectures (CNN, LSTM, Transformer) made training stable and effective.
- ***Open-source tools and community:*** Libraries such as scikit-learn, TensorFlow and PyTorch, and open datasets and pre-trained models, lowered the entry barrier and sped up research.
- In the 1960s-80s, ideas like neural networks existed but lacked data and compute; in the 2010s, their availability turned the same ideas into practical success.

Q6. Discuss the major issues and challenges in Machine Learning, including insufficient or poor-quality data, noisy data, overfitting, underfitting, data leakage, class imbalance, and computational constraints.

Challenge	***Meaning and Effect***	***Common Remedies***
Insufficient data	Too few examples to capture the true pattern; the model cannot generalise and gives unstable results.	Collect more data, data augmentation, transfer learning, synthetic data, simpler models.
Poor-quality data	Data that is inaccurate, incomplete, inconsistent, biased or irrelevant ("garbage in, garbage out"); it leads to wrong or biased models.	Data cleaning, imputation of missing values, validation rules, better collection processes, bias audit.
Noisy data	Random errors or outliers in features or labels (sensor errors, wrong labels) that hide real patterns and make the model learn the noise.	Outlier detection, smoothing, robust algorithms, label verification, regularisation.
Overfitting	Model performs very well on training data but poorly on new data because it has memorised noise and details. Indicates high variance.	More data, simpler model, regularisation (L1/L2), dropout, early stopping, pruning, cross-validation, ensembling.
Underfitting	Model is too simple to capture the pattern and performs poorly on both training and test data. Indicates high bias.	More complex model, better/more features, longer training, reducing regularisation.
Data leakage	Information from outside the training set (test data, or future information / the target itself) accidentally enters training, giving unrealistically high accuracy that fails in production.	Split data before preprocessing, fit scalers only on training data, use time-based splits, remove features unavailable at prediction time.
Class imbalance	One class has far fewer examples than the others (fraud, rare disease); the model favours the majority class, and accuracy becomes misleading.	Resampling (SMOTE, undersampling), class weights, suitable metrics (precision, recall, F1, PR-AUC), anomaly detection methods.
Computational constraints	Training and running large models needs much memory, processing time, energy and money; real-time or edge devices have strict limits.	Efficient algorithms, feature selection, dimensionality reduction, model compression (pruning, quantisation), GPUs / cloud, distributed training.

Bias-Variance Trade-off

Overfitting and underfitting are two ends of the ***bias-variance trade-off***. A model with high bias is too simple (underfits), while a model with high variance is too sensitive to the training data (overfits). The aim is to find the right complexity so that total error on unseen data is minimum. ***Other challenges*** include lack of interpretability of complex models, bias and fairness, privacy and security, concept drift after deployment, and shortage of skilled people.

Q7. Differentiate between Data Science and Machine Learning with respect to objectives, scope, techniques, workflows, and career roles. Explain how Machine Learning fits within the broader Data Science lifecycle.

Basis	***Data Science***	***Machine Learning***
Objective	To extract knowledge and insights from data to support decisions and solve problems.	To build models that learn from data to make predictions or decisions automatically.
Scope	Broad, multidisciplinary field: data collection, cleaning, analysis, visualisation, modelling, communication and deployment.	Narrower; a subfield of AI and a key tool within data science, concentrating on learning algorithms.
Techniques	Statistics, data mining, SQL, data wrangling, visualisation, ML, big data tools.	Regression, classification, clustering, neural networks, optimisation, model tuning and evaluation.
Workflow	Business understanding, data acquisition, cleaning, EDA, modelling, evaluation, deployment, communication.	Feature preparation, algorithm selection, training, validation, tuning, testing, and deployment of the model.
Output	Reports, dashboards, insights, recommendations, and predictive models.	A trained model or algorithm integrated into a product.
Data type	Structured, semi-structured and unstructured data.	Mostly prepared numerical / encoded data for training.
Tools	Python, R, SQL, Excel, Tableau / Power BI, Spark.	scikit-learn, TensorFlow, PyTorch, Keras, XGBoost.
Career roles	Data Scientist, Data Analyst, Business Intelligence Analyst, Data Engineer.	Machine Learning Engineer, AI Research Scientist, Deep Learning Engineer, MLOps Engineer, NLP / Computer Vision Engineer.

Place of Machine Learning in the Data Science Lifecycle

1. ***Business understanding:*** define the problem and success criteria.
2. ***Data acquisition and understanding:*** collect data from various sources.
3. ***Data preparation:*** cleaning, integration, transformation.
4. ***Exploratory data analysis:*** discover patterns, correlations and anomalies.

5. ***Modelling (Machine Learning stage):*** feature engineering, model training and tuning using ML algorithms. This is where ML is applied.
6. ***Evaluation:*** check results against business objectives and metrics.
7. ***Deployment and monitoring:*** put the model into production and track its performance.
8. ***Communication and decision making:*** present insights and actions to stakeholders.

Thus ML is the ***modelling engine*** of the data science lifecycle. Not every data science project needs ML (some only need analysis and reporting), but ML projects always depend on the data science steps before and after modelling.

Q8. Consider a university that wants to predict whether students are at risk of dropping out. Formulate this as a Machine Learning problem by identifying the input data, target variable, learning type, possible evaluation metrics, and major ethical or privacy concerns. Problem Statement

Given the information available about a student up to a certain point (for example, the end of the first semester), predict whether the student is at risk of dropping out, so that the university can give timely help such as counselling, mentoring or financial support.

1. Input Data (Features)

- ***Academic data:*** marks / grades and CGPA trend, backlogs or failed subjects, attendance percentage, assignment and exam performance, admission test score, previous board marks.
- ***Engagement data:*** learning management system (LMS) logins, submission timeliness, library usage, participation in classes and activities.
- ***Financial data:*** fee payment delays, scholarship or loan status.
- ***Demographic and background data*** (used with great care): age, distance from home, hostel or day scholar, first-generation learner, part-time work.
- ***Support-related data:*** counselling visits, mentor meetings, disciplinary records.

2. Target Variable

A binary label, ***Dropout (1) / Not dropout (0)***, defined as whether the student left the programme (or did not re-register) within a fixed period, e.g., the next 1-2 semesters. The labels come from historical records of previous batches. Alternatively, a multi-class risk level (low / medium / high) can be used.

3. Learning Type

Supervised learning - binary classification, since historical data with known outcomes (dropped out or continued) is available. Suitable algorithms are logistic regression (interpretable baseline), decision tree, random forest and gradient boosting. Unsupervised clustering can additionally be used to identify groups of students with similar risk patterns.

4. Evaluation Metrics

- ***Class imbalance*** exists (dropouts are usually a small minority), so accuracy alone is misleading.

- ***Recall (sensitivity)*** for the at-risk class: the percentage of actual dropouts correctly identified; missing at-risk students is costly.
- ***Precision:*** the percentage of students flagged as at-risk who are really at risk; it avoids wasting counselling resources.
- ***F1-score***, ***ROC-AUC*** and ***PR-AUC*** for overall performance, and the ***confusion matrix*** for error types.
- ***Calibration*** of predicted probabilities and ***precision at top-k*** (accuracy among the k students with highest risk scores).
- ***Fairness metrics:*** compare error rates across groups such as gender, income or region.
- Use of a ***time-based train / test split*** (train on older batches, test on the latest) and cross-validation, to avoid leakage and to reflect real use.

5. Ethical and Privacy Concerns

- ***Privacy and consent:*** student data is personal and sensitive. Data should be collected with informed consent, kept minimal (data minimisation), stored securely, access-controlled, and used only for the stated purpose, in line with data protection laws such as India's Digital Personal Data Protection Act, 2023.
- ***Bias and fairness:*** historical data may reflect discrimination; the model may unfairly label students from certain gender, caste, economic or regional groups as high risk. Sensitive attributes and proxies must be examined and audited.
- ***Stigma and self-fulfilling prophecy:*** labelling a student "high risk" may lower teachers' expectations or affect the student's confidence. Predictions should be confidential and used only to offer support, never for punishment or exclusion.
- ***Transparency and explainability:*** the university should be able to explain why a student was flagged; students should know that such a system is used.
- ***Human oversight:*** the prediction is one input, and advisers make the final decision. Errors (false positives and false negatives) must be handled responsibly.
- ***Data security and retention:*** protection from breaches and deletion of data when no longer required.

Element	***Formulation***
Input	Academic, engagement, financial and background features of each student.
Target	Dropout within next 1-2 semesters (1 = yes, 0 = no).
Learning type	Supervised binary classification.
Metrics	Recall, precision, F1, PR-AUC / ROC-AUC, calibration, fairness checks.
Ethics	Consent, privacy, bias, transparency, human oversight, non-punitive use.

Q9. Compare classification, regression, clustering, and reinforcement learning in terms of the type of problem they solve, expected output, and examples of practical use. Give one suitable example for each.

Basis	***Classification***	***Regression***	***Clustering***	***Reinforcement Learning***
Learning paradigm	Supervised.	Supervised.	Unsupervised.	Learning by interaction and rewards.
Problem solved	Assign an input to one of the predefined categories.	Predict a continuous numerical value.	Group similar data points when no labels are given.	Choose a sequence of actions in an environment to maximise cumulative reward.
Expected output	Discrete class label (or class probabilities).	Real number.	Cluster assignments / groups.	A policy (best action for each state).
Data needed	Labelled data with categories.	Labelled data with numeric targets.	Unlabelled data.	Environment with reward signal; states, actions, rewards.
Common algorithms	Logistic regression, decision tree, SVM, k-NN, Naive Bayes.	Linear / polynomial regression, ridge, lasso, regression trees.	k-means, hierarchical, DBSCAN.	Q-learning, DQN, policy gradient.
Evaluation	Accuracy, precision, recall, F1.	MAE, MSE, RMSE, R^2.	Silhouette score, Davies-Bouldin index.	Total / average reward.
Practical uses	Spam filtering, disease diagnosis, sentiment analysis.	House price, sales, weather, stock trend forecasting.	Customer segmentation, document grouping, image compression.	Robotics, games, autonomous driving, ad bidding.
Example	***Email spam detection:*** classify an email as spam or not spam from its words and sender.	***House price prediction:*** predict price from area, location and number of rooms.	***Customer segmentation:*** group shoppers by purchase behaviour for targeted marketing.	***Robot navigation / game agent:*** a robot learns to reach a target while avoiding obstacles, gaining reward for progress and penalty for collisions.

Q10. Critically discuss the statement: "A high-performing Machine Learning model is not necessarily a successful Machine Learning system." Explain the technical and practical factors that should be considered beyond model accuracy.

The statement is ***largely true***. A model is only one component of an ML system. A model that scores very well on a test set may still fail to deliver value once it is placed in the real world, because a ***system*** also includes data pipelines, infrastructure, users, business processes, monitoring and people. Success means the system reliably solves the real problem, safely and at acceptable cost, over time.

Why Accuracy Alone Is Not Enough

Accuracy can be misleading. In a fraud dataset with 1% fraud, a model that predicts "genuine" for every transaction is 99% accurate, but completely useless. Test data may also not represent real-world data, so good offline results may not carry over to production. A well-known illustration is the Netflix Prize: the winning ensemble improved the prediction error, but Netflix reported that it was not deployed because the engineering effort needed was not justified by the gain.

Technical Factors Beyond Accuracy

- ***Generalisation and robustness:*** performance on unseen, noisy or changing data and on edge cases; resistance to adversarial inputs.
- ***Suitable metrics:*** precision, recall, F1, calibration and cost-sensitive measures aligned with the real cost of errors.
- ***Data quality and drift:*** the input data distribution (data drift) and the relationship between inputs and output (concept drift) change over time, degrading performance unless monitored and retrained.
- ***Data leakage and validation:*** unrealistic offline results due to leakage or incorrect evaluation.

- ***Interpretability and explainability:*** ability to explain predictions (e.g., using SHAP, LIME or simple models), essential in healthcare, finance and law.
- ***Fairness and bias:*** equitable performance across different groups.
- ***Latency, scalability and efficiency:*** prediction speed, memory and compute requirements, and ability to handle many users.
- ***Reproducibility, versioning and MLOps:*** tracking of data, code and models; automated testing, deployment (CI/CD), monitoring and rollback.
- ***Security and privacy:*** protection against data leakage, model theft and poisoning.

Practical Factors Beyond Accuracy

- ***Business value and ROI:*** does the system solve a real problem and produce measurable benefit that exceeds development and running costs?
- ***Integration with existing systems and workflows:*** the model must fit into the organisation's software and processes.
- ***User trust and adoption:*** if users do not understand or trust the output, they will not use it.
- ***Ethical, legal and regulatory compliance:*** data protection laws, industry regulations, accountability and social impact.

- ***Maintainability and technical debt:*** ML systems need continuous data management, retraining and monitoring; long-term maintenance cost is often larger than the initial development cost.
- ***Feedback loops:*** predictions can influence future data (e.g., a recommender showing only popular items), amplifying bias.
- ***Human oversight and fail-safe design:*** what happens when the model is wrong or unavailable; human-in-the-loop for critical decisions.
- ***Stakeholder communication and team skills:*** collaboration between data scientists, engineers, domain experts and management.